MIL-D-14258A(MU)
1 April 1968
SUPERSEDING
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17 February 1956

MILITARY SPECIFICATION

DRUM, FIBER, 20-GALLON CAPACITY, FOR CHEMICALS

1. SCOPE

1.1 This specification covers one type of fiber drum for use in shipping chemical agents.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Box, Wood, Cleated-Plywood.
PPP-B-636 - Box, Fiberboard.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 101 - Preservation, Packaging, and Packing Material: Test Procedures.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

C6-16-1 - Drum, Fiber, 20-Gallon Capacity - Assembly, Detail, and Bill of Material.

FSC 8110

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Material and components. All materials and components used in the manufacture and assembly of the drum shall conform to the specifications and drawings forming a part of this specification.

3.2 Manufacture and assembly. The drum shall be manufactured and assembled in accordance with Drawing (Dwg) C6-16-1 and subsidiary drawings listed thereon.

3.3 Water resistance. When tested as specified in 4.4.4.1, the sealed, empty drum shall show no evidence of warpage, delamination, or mositure inside the drum.

3.4 Compression strength.

3.4.1 Dry strength. When tested dry as specified in 4.4.4.2, the sealed, empty drum shall not fail at any value less than 4500 pounds. Minor deformation, other than that in the drum body, shall not be cause for rejection.

3.4.2 Wet strength. When tested wet as specified in 4.4.4.2, the sealed, empty drum shall not fail at any value less than 4000 pounds. Minor deformation, other than in the drum body, shall not be cause for rejection.

3.5 Bursting strength.

3.5.1 Filled container. When tested as specified in 4.4.4.3, the filled drum shall not burst, leak, break or become excessively deformed.

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3.5.2 Dry component specimens. When tested as specified in 4.4.4.4, dry specimens from the drum components shall meet the dry bursting strengths specified in table I.

3.5.3 Wet component specimens. When tested as specified in 4.4.4.4, wet specimens from the drum components shall meet the wet bursting strengths specified in table I.

Table I. Bursting strengths of components (psi)

Condition	Cover heading	Outer tube of cover	Cover pad	Inner tube of shell	Outer tube of shell	Bottom shell pad	Bottom heading
Dry	950	600	350	450	575	600	950
Wet	400	125	50	60	160	90	260

3.6 Preproduction. Unless otherwise specified in the contract or order, a preproduction lot of drums shall be produced in accordance with the requirements stated herein, for examination and tests (see 4.3).

4. QUALTIY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

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4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 100 drums shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination. Each drum in the preproduction sample shall be inspected for conformance to the drawing and specification requirements cited herein and in accordance with the classification of defects.

4.3.2.2 For tests. Sample drums for each consecutive test shall be randomly selected from those remaining in the preproduction sample.

(a) Water resistance. Forty-four drums shall be tested as specified in 4.4.4.1.

(b) Compression strength. Twenty-two drums shall be dry tested as specified in 4.4.4.2. Twenty-two drums, randomly selected from the 44 tested in (a) above, shall be wet tested as specified in 4.4.4.2.

(c) Bursting strength-filled container. Twenty-two drums shall be tested as specified in 4.4.4.3.

(d) Bursting strength-components. Upon completion of the compression strength tests, ten drums shall be selected from each 22-drum sample. One specimen of each component listed in table I, 6 inches by 6 inches in size, shall be taken from each sample drum, providing ten dry and ten wet specimens of each component. The specimens shall be tested as specified in 4.4.4.4.

4.3.3 Acceptance/rejection criteria. Failure of any drum in the preproduction lot to meet the requirements of any test shall result in rejection of the lot.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the drums produced by one manufacturer, in not more than one calendar week, from the same materials, and under essentially the same manufacturing conditions.

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4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For tests. Sampling shall be conducted in accordance with MIL-STD-105, level S-2.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample drums and levels A and B preparation for delivery thereof shall be examined in accordance with MIL-STD-105 and the classification of defects.

4.4.3.2 For tests.

(a) Water resistance. Sample drums shall be tested in accordance with 4.4.4.1 and with MIL-STD-105, level S-2, and an AQL of 1.0 percent defective.

(b) Compression strength. Sample drums shall be tested in accordance with 4.4.4.2 and with MIL-STD-105, level S-2, and an AQL of 1.5 percent defective. One half of the drums in the sample shall be dry tested; the remainder shall be wet tested.

(c) Bursting strength-filled drum. Sample drums shall be tested in accordance with 4.4.4.3 and with MIL-STD-105, level S-2, and an AQL of 1.5 percent defective.

(d) Bursting strength-components. Sample drums shall be tested in accordance with MIL-STD-105, level S-2, and an AQL of 1.5. The same sample selected for the compression strength test (see 4.4.3.2(b) may be used. Half of the drums in the sample shall be dry tested; the other half shall be wet tested. Three specimens, 6 inches by 6 inches, of each component listed in table I shall be taken from each drum in the sample. The specimens shall be tested in accordance with 4.4.4.4.

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4.4.3.3 Classification of defects. Sample drums and the levels A and B preparation for delivery thereof shall be examined as follows:

(a) Drum, fiber, 20-gallon assembly (Dwg C6-16-1).

Categories	Defects
Critical:	None defined
Major:	AQL 1.5 percent defective
101	Inside diameter incorrect
102	Length incorrect
103	Drum damaged or contaminated
104	Metal stitches missing or deformed
105	Resin and paraffin mix missing or inadequate
106	Pads missing, incorrect or incorrectly attached (bottom pad)
107	Unglued area of outer ply greater than 25 percent
108	Outer tube and cover not convolutely wound
109	Outer ply of outer tube not fabricated from asphalt laminated paper
110	Packing container incorrect
Minor:	AQL 2.5 percent defective
201	Marking missing, illegible, or incorrect
202	Drum cover does not fit properly
203	Zinc oxide coating missing
204	Tape missing or inadequate
205	Heading incorrect or incorrectly attached

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Water resistance. Close and seal the drum in accordance with Dwg C6-16-1. Subject the drum to a continuous water spray for 24 hours. The water temperature shall be 73° ± 2°F. The spray nozzles shall be no less than 6 feet above the top of the drum and shall deliver the water as a heavy mist of small droplets of uniform concentration. Rate of water flow of the spray shall be 4 ± 0.8 inch per hour, measured at the top surface of the drum. When more than one drum is to be tested, arrange the drums in a single layer, in a vertical position, with the drums spaced no less than 6 inches apart. After the 24 hours of continuous water spray, examine the drum or drums for conformance to 3.3.

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4.4.4.2 Compression strength. Close and seal the drum in accordance with Dwg C6-16-1. Apply a dynamic top compression load at a rate of 9/16 inch per minute. The dry test drum shall meet the requirement of 3.4.1 and the wet test drum shall meet the requirement of 3.4.2.

4.4.4.3 Bursting strength-filled drum. Fill the drum with an intimate mixture of sand and sawdust having a density of 5 pounds per gallon. Close and seal the drum in accordance with Dwg C6-16-1. Drop the drum from a height of 4 feet ± 2 inches onto a solid concrete floor, so as to strike diagonally on the top and then diagonally on the bottom. The drum shall meet the requirements of 3.5.1.

4.4.4.4 Bursting strength-components. Bursting strengths of the component specimens shall be determined in accordance with method 250 of Fed. Test Method Std. No. 101. Prepare the wet component specimens by immersing them in tap water at 73° ± 3°F for 24 hours, then blotting dry until the specimen surfaces no longer glisten. The average bursting strength of the dry specimens of each component type and of the wet specimens of each component type shall meet the bursting strengths specified for their respective conditions in table I.

5. PREPARATION FOR DELIVERY

5.1 Packing. Drums shall be packed level A, B, or C, as specified (see 6.2).

5.1.1 Level A. The top pad assembly shall be placed loosely in each drum, assuring that the coated Kraft paper sealing disk is not torn. The slip cover shall then be placed in position on each drum. Four drums so prepared shall be packed upright in a cleated-plywood box conforming to a type overseas, style A or B box for a type 2 average load of PPP-B-601. The filled box shall be closed and strapped with galvanized strapping, as specified in the appendix to the box specification.

5.1.2 Level B. The drums shall be packed as specified in 5.1.1 using a box conforming to style RSC, type CF, class domestic, variety SW, grade 350 for a type 2 average load of PPP-B-636. Each box shall be closed in accordance with the closure requirements of method II of the box specification and shall be reinforced using tape banding as specified in the appendix to PPP-B-636. The dimensional limitations in table V of the box specification shall be waived.

5.1.3 Level C. Drums shall be packed to provide adequate protection against deterioration and damage and to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall comply with Uniform Freight Classification Rules or the regulations of other common carriers applicable to the mode of transportation.

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5.2 Marking. In addition to the marking specified in Dwg C6-16-1 and to any special marking specified in the contract or order, (see 6.2) shipping containers shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The fiber drum covered by this specification is intended for use as a container for chemical agents such as impregnite (CC2 and XXCC3).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packing required.
(c) Special marking other than specified.
(d) Deletion of preproduction lot, if desired.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 8110-A139

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